MERCK MANUAL
Professional Version

Dementia

(Major Neurocognitive Disorder)

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Dementia is chronic, global, usually irreversible deterioration of cognition. Diagnosis is clinical; laboratory and imaging tests are used to identify treatable causes. Treatment is supportive. Medications tailored to the specific type of dementia can sometimes improve cognitive function.

(See also Overview of Delirium and Dementia.)

Dementia may occur at any age but primarily affects older adults. It accounts for more than half of nursing home admissions.

Dementia can be classified in several ways:

- Alzheimer or non-Alzheimer type
- Cortical or subcortical
- Irreversible or potentially reversible

Dementia should not be confused with delirium, although cognition is disordered in both. The following helps distinguish them:

- **Dementia** affects mainly memory, is typically caused by anatomic changes in the brain, has slower onset, and is generally irreversible.
- **Delirium** affects mainly attention, is typically caused by acute illness or drug toxicity (sometimes life threatening), and is often reversible.

Other specific characteristics also help distinguish the two disorders (see table Differences Between Delirium and Dementia).

Etiology of Dementia

Dementias may result from primary diseases of the brain or other conditions (see table Classification of Some Dementias).

The most common types of dementia are

- Alzheimer disease
- Vascular dementia
- Dementia with Lewy bodies
- Frontotemporal dementia

Dementia also occurs in patients with Parkinson disease, Huntington disease, progressive supranuclear palsy, Creutzfeldt-Jakob disease, Gerstmann-Sträussler-Scheinker syndrome, other prion disorders, neurosyphilis, HIV infection, a traumatic brain injury (eg, chronic traumatic encephalopathy), or certain brain tumors located in cortical or subcortical brain areas involved in cognition. Patients can have > 1 type (mixed dementia). The most common type of mixed dementia is Alzheimer disease combined with vascular dementia.

Some structural brain disorders (eg, normal pressure hydrocephalus, subdural hematoma), metabolic disorders (eg, hypothyroidism, vitamin B12 deficiency), and toxins (eg, lead) cause a slow deterioration of cognition that may resolve with treatment. This impairment is sometimes called reversible dementia, although some experts restrict the term "dementia" to irreversible cognitive deterioration.

Depression may mimic dementia (and was formerly called pseudodementia); the 2 disorders often coexist. However, depression may be the first manifestation of dementia.

Age-associated memory impairment refers to changes in cognition that occur with aging. Older adults have a relative deficiency in recall, particularly in speed of recall. These changes do not affect daily functioning and thus do not indicate dementia. However, the earliest manifestations of dementia are very similar.

Mild cognitive impairment (MCI) causes greater memory loss than age-associated memory impairment; memory and sometimes other cognitive functions are worse in patients with MCI than in age-matched controls, but daily functioning is typically not affected. In contrast, dementia impairs daily functioning. Up to 50% of patients with mild cognitive impairment develop dementia within 3 years.

Subjective cognitive decline (SCD) is defined as a self-experienced persistent decline in cognitive capacity but normal performance on standardized cognitive tests used to classify MCI (1). The risk of MCI and dementia is increased in people with SCD.

Various disorders may exacerbate cognitive deficits in patients with dementia. Delirium often occurs in patients with dementia.

Medications, particularly benzodiazepines and anticholinergics (eg, some tricyclic antidepressants, antihistamines, antipsychotics, benztropine), may temporarily cause or worsen symptoms of dementia, as may alcohol or recreational drugs, even in moderate amounts. New or progressive kidney or liver failure may reduce medication clearance and cause toxicity after years of taking a stable dose (eg, of propranolol).

Prion-like propagation mechanisms appear to be involved in most or all neurodegenerative disorders that first manifest in older patients. A normal cellular protein, either sporadically or via an inherited mutation, becomes misfolded into a pathogenic form or prion. The prion then acts as a template, causing other proteins to misfold similarly. This process occurs over years and in many parts of the central nervous system (CNS). Many of these prions become insoluble and, like amyloid, cannot be readily cleared by the cell. Evidence supports prion or similar mechanisms in Alzheimer disease (2), as well as in Parkinson disease, Huntington disease, frontotemporal dementia, and amyotrophic lateral sclerosis. These prions are not as infectious as those in Creutzfeld-Jacob disease, but they can be transmitted.

Rapidly progressive dementia (RPD) is a group of heterogeneous cognitive disorders that progress faster than other dementia syndromes, typically within 1 to 2 years (3). The most prominent presentation is cognitive decline (eg, memory loss, visuospatial and language deficits, executive dysfunction). However, other neuropsychiatric symptoms (eg, behavior disturbance, personality change, mood disorders, psychosis, sleep disturbance, altered alertness and/or awareness, involuntary movements such as tremor and myoclonus, gait disturbance, seizure-like activities, ataxia, parkinsonian features) also occur. Prion disease is the most common cause of RPD. Other common causes include autoimmune and paraneoplastic encephalitis. The progression of other dementias may also be atypically rapid; they include Alzheimer disease, dementia with Lewy bodies, frontotemporal dementia, and dementia due to potentially reversible causes (eg, infectious, toxic/metabolic, neurovascular, psychiatric).

TABLE

Classification of Some Dementias

Classification	Examples
Beta-amyloid deposits and neurofibrillary tangles	Alzheimer disease
Tau abnormalities	Chronic traumatic encephalopathy Corticobasal ganglionic degeneration Frontotemporal dementia (including Pick disease) Progressive supranuclear palsy
Alpha-synuclein abnormalities	Dementia with Lewy bodies Parkinson disease dementia
Autoimmune disorders	Autoimmune dementia, including postencephalitis syndromes
Huntingtin gene mutation	Huntington disease

Classification	Examples
Cerebrovascular disease	Vascular dementias: • Binswanger disease • Lacunar disease • Multi-infarct dementia • Strategic single-infarct dementia • Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL) and cerebral autosomal recessive arteriopathy with subcortical infarcts and leukoencephalopathy (CARASIL)
Ingestion of drugs or toxins	Alcohol-associated dementia Dementia due to exposure to heavy metals
Infections	**Fungal:** Dementia due to cryptococcosis **Spirochetal:** Dementia due to syphilis or Lyme disease **Viral:** HIV-associated dementia
Paraneoplastic disorders	Paraneoplastic neurologic syndrome with dementia
Prion disorders	Alzheimer disease Creutzfeldt-Jakob disease Frontotemporal dementia Huntington disease Parkinson disease Variant Creutzfeldt-Jakob disease
Structural brain disorders	Brain tumor Chronic subdural hematoma Normal-pressure hydrocephalus White matter disorders: Adult-onset leukodystrophies, multiple sclerosis
Other potentially reversible disorders	Depression

Classification	Examples
	Hypothyroidism Vitamin B12 deficiency

Etiology references

1. Jessen F, Amariglio RE, Buckley RF, et al: The characterisation of subjective cognitive decline. *Lancet Neurol* 19 (3):271–278, 2020. doi: 10.1016/S1474-4422(19)30368-0
2. Crestini A, Santilli F, Martellucci S, et al: Prions and Neurodegenerative Diseases: A Focus on Alzheimer's Disease. *J Alzheimers Dis* 85(2):503–518, 2022. doi:10.3233/JAD-215171
3. Hermann P, Zen I: Rapidly progressive dementias — aetiologies, diagnosis and management. *Nat Rev Neurol* 18 (6):363–376, 2022. doi: 10.1038/s41582-022-00659-0

Symptoms and Signs of Dementia

(See also Behavioral and Psychologic Symptoms of Dementia.)

Dementia impairs cognition globally. Onset is gradual, although family members may suddenly notice deficits (eg, when function becomes impaired). Often, loss of short-term memory is the first sign. At first, early symptoms may be indistinguishable from those of age-associated memory impairment or mild cognitive impairment, but then progression becomes apparent.

Although symptoms of dementia exist on a continuum, they can be divided into

- Early
- Intermediate
- Late

Personality changes and behavioral disturbances may develop early or late. Motor and other focal neurologic deficits occur at different stages, depending on the type of dementia; they occur early in vascular dementia and late in Alzheimer disease. Incidence of seizures is somewhat increased during all stages.

Psychosis, presenting as hallucinations, delusions, or paranoia, can occur in approximately 20% to 50% of patients with dementia, depending on the type of dementia and disease severity (1).

Many concurrent general medical disorders, particularly during exacerbations, can worsen the symptoms of dementia. These disorders include diabetes, chronic bronchitis, emphysema, infections, chronic kidney disease, liver disorders, and heart failure.

Drinking alcohol, even in moderate amounts, may also worsen the symptoms of dementia and contribute to its progression (eg, by reducing brain volume); most experts recommend that patients with dementia stop drinking alcohol.

Early (mild) dementia symptoms

Recent memory is impaired; learning and retaining new information become difficult. Language problems (especially with word finding), mood swings, and personality changes develop. Patients may have progressive difficulty with independent activities of daily living (eg, balancing their checkbook, finding their way around, remembering where they put things). Abstract thinking, insight, or judgment may be impaired. Patients may respond to loss of independence and memory with irritability, hostility, and agitation.

Functional ability may be further limited by the following:

- Agnosia: Impaired ability to identify objects despite intact sensory function
- Apraxia: Impaired ability to do previously learned motor activities despite intact motor function
- Aphasia: Impaired ability to comprehend or use language

Although early dementia may not compromise sociability, family members may report strange behavior accompanied by emotional lability.

Intermediate (moderate) dementia symptoms

Patients become unable to learn and recall new information. Memory of remote events is reduced but not totally lost. Patients may require help with basic activities of daily living (eg, bathing, eating, dressing, toileting).

Personality changes may progress. Patients may become irritable, anxious, self-centered, inflexible, or more readily angry. They may become more passive, with a flat affect; they may develop depression, become indecisive, lose spontaneity, or generally withdraw from social situations. Personality traits or habits may become more exaggerated (eg, concern with money becomes obsession with it).

Behavior disorders may develop: Patients may wander or become suddenly and inappropriately agitated, hostile, uncooperative, or physically aggressive.

By this stage, patients have lost all sense of time and place because they cannot effectively use normal environmental and social cues. Patients often get lost; they may be unable to find their own bedroom or bathroom. They remain ambulatory but are at risk of falls or accidents secondary to confusion.

Altered sensation or perception may culminate in psychosis with hallucinations and paranoid and persecutory delusions.

Sleep patterns are often disorganized.

Late (severe) dementia symptoms

Patients cannot walk, feed themselves, or do other activities of daily living; they may become incontinent. Recent and remote memory is completely lost. Patients may be unable to swallow. They are at risk of undernutrition, pneumonia (especially due to aspiration), and pressure ulcers. Because they depend completely on others for care, placement in a long-term care facility often becomes necessary. Eventually, patients become mute.

Because such patients cannot relate symptoms to a clinician and because older patients often have diminished febrile or leukocytic response to infection, clinicians must rely on experience and acumen whenever a patient appears ill.

End-stage dementia results in coma and death, usually due to infection.

Symptoms and signs reference

1. Cressot C, Vrillon A, Lilamand M, et al: Psychosis in Neurodegenerative Dementias: A Systematic Comparative Review. *J Alzheimers Dis* 99(1):85–99, 2024. doi:10.3233/JAD-231363

Diagnosis of Dementia

- Differentiation of delirium from dementia by history and neurologic examination (including mental status)
- Identification of treatable causes, based on clinical evaluation and selected laboratory testing and neuroimaging
- Sometimes formal neuropsychological testing

Distinguishing the type or cause of dementia can be difficult; definitive diagnosis often requires postmortem pathologic examination of brain tissue. Thus, clinical diagnosis focuses on distinguishing dementia from delirium and other disorders and identifying the cerebral areas affected and potentially reversible causes.

Dementia must be distinguished from the following:

- **Delirium:** Distinguishing between dementia and delirium can be difficult but is crucial because delirium is usually reversible with prompt treatment. Attention is assessed first. Inattention is most consistent with the diagnosis of delirium, although advanced dementia also severely impairs attention. Other features that suggest delirium (eg, short duration of cognitive impairment) are determined by the history, physical examination, and tests for specific causes.
- **Age-associated memory impairment:** Memory impairment does not affect daily functioning. If affected people are given enough time to learn new information, their intellectual performance is good.
- **Mild cognitive impairment:** Memory and/or other cognitive functions are impaired, but impairment is not severe enough to interfere with daily activities.
- **Cognitive symptoms related to depression:** This cognitive disturbance resolves with treatment of depression. Depressed older patients may experience cognitive decline, but unlike patients with dementia, they tend to exaggerate their memory loss and rarely forget important current events or personal matters. Neurologic examination is normal except for

signs of psychomotor slowing. When tested, patients with depression make little effort to respond, while those with dementia often try hard but respond incorrectly. When depression and dementia coexist, treating depression does not fully restore cognition.

Clinical criteria

There are various clinical criteria available to help diagnose dementia, and the specific subtypes (1). One of the most commonly used systems includes the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5-TR). The DSM-5-TR classifies dementia under the broader category of neurocognitive disorders, which can be further subdivided depending on the presumed etiology (eg, neurodegenerative diseases like Alzheimer's disease). The diagnosis of dementia is based primarily on cognitive, behavioral, and functional symptoms (2).

To meet the criteria, patients have evidence of significant cognitive decline from a previous level of performance in ≥ 1 of the following cognitive domains:

- Complex attention
- Executive function
- Learning and memory
- Language
- Perceptual-motor cognition
- Social cognition

Evidence of cognitive decline may come from the individual, or an informed contact or clinician, and the impairment has been documented, preferably by standardized neuropsychological testing, or if unavailable, another quantified clinical assessment.

In addition, the cognitive deficits interfere with independence in daily activities, do no occur exclusively in the context of a delirium, and are not better explained by another mental disorder (eg, major depressive disorder, schizophrenia).

If cognitive impairment is confirmed, history and physical examination should then focus on signs of treatable disorders that cause cognitive impairment (eg, vitamin B12 deficiency, neurosyphilis, hypothyroidism, depression—see table Causes of Delirium).

Assessment of cognitive function

The **Mini-Mental Status Examination** (see sidebar Examination of Mental Status) or the Montreal Cognitive Assessment (MoCA), is often used as a bedside screening test (3). When delirium is absent, the presence of multiple deficits, suggests dementia. The best screening test for memory is a short-term memory test (eg, registering 3 objects and recalling them after 5 minutes); patients with dementia fail this test. Another test of mental status assesses the ability to name multiple objects within categories (eg, lists of animals, plants, or pieces of furniture). Patients with dementia struggle to name a few; those without dementia easily name many. The patient's educational background must be factored in when interpreting cognitive test results. Patients with fewer years of education tend to score lower on cognitive tests, whch can lead to potential misdiagnosis of cognitive impairment.

Neuropsychologic testing should be done when history and bedside mental status testing are not conclusive. It evaluates mood as well as multiple cognitive domains. Testing is performed or supervised by a neuropsychologist and may take 1 to 3 hours to complete. Such testing helps primarily in differentiating the following:

- Age-associated memory impairment, mild cognitive impairment, and dementia, particularly when cognition is only slightly impaired or when the patient or family members are anxious for reassurance
- Dementia and focal syndromes of cognitive impairment (eg, amnesia, aphasia, apraxia, visuospatial difficulties, impairment of executive function) when the distinction is not clinically evident

Testing may also help characterize specific deficits due to dementia, and it may detect depression or a personality disorder that is contributing to poor cognitive performance.

Laboratory testing

Tests should include thyroid-stimulating hormone (TSH) and vitamin B12 levels. Routine complete blood count (CBC) and kidney and liver function tests are sometimes recommended, but their yield for identifying underlying disease is very low.

If clinical findings suggest a specific disorder, other tests (eg, for HIV or syphilis) are indicated. Lumbar puncture should be considered if chronic infection or neurosyphilis is suspected or in patients with a rapidly progressive dementia to evaluate for possible prion disease or an autoimmune disorder.

Biomarker testing may be useful in selected patients with suspected Alzheimer disease. For example, elevated phosphorylated tau level and decreased beta-amyloid level in the cerebrospinal fluid (CSF) are considered a signature change for Alzheimer disease.

Routine genetic testing for the presence of epsilon-4 alleles (ApoE4) is not recommended as part of the diagnostic work-up for dementia. Although data suggest that the *apolipoprotein E* gene increases the risk of developing Alzheimer disease (4), it is neither necessary nor sufficient to cause Alzheimer disease.

Neuroimaging

CT or MRI should be done in the initial evaluation of dementia and after any unexplained change in cognition or mental status. Neuroimaging can identify potentially reversible structural disorders (eg, normal pressure hydrocephalus, brain tumors, subdural hematoma) and certain metabolic disorders (eg, pantothenate kinase-associated neurodegeneration, Wilson disease) and irreversible disorders (eg, stroke, leukodystrophy, Creutzfeld-Jacob disease).

Occasionally, electroencephalography (EEG) is useful (eg, to evaluate episodic lapses in attention or bizarre behavior).

PET with fluorine-18 (18F)–labeled deoxyglucose (fluorodeoxyglucose, or FDG) or single-photon emission CT (SPECT) can provide information about cerebral perfusion patterns and help differentiate Alzheimer disease from frontotemporal dementia and dementia with Lewy bodies (5).

Amyloid radioactive tracers that bind specifically to beta-amyloid plaques (eg, fluorine-18 [18F] florbetapir, [18F] flutemetamol, [18F] florbetaben) have been used with positron emission tomography (PET) to image amyloid plaques in patients with mild cognitive impairment or dementia. This testing should be used when the cause of cognitive impairment (eg, mild cognitive impairment or dementia) is uncertain after a comprehensive evaluation and when Alzheimer disease is a diagnostic consideration. Determining amyloid status via PET is expected to increase the certainty of diagnosis and management. [18F] flortaucipir-PET, using a tau radioactive tracer, can be used to estimate the density and distribution of aggregated tau neurofibrillary tangles in adults who have cognitive impairment and are being evaluated for Alzheimer disease (5).

Diagnosis references

1. NICE: National Institute for Health and Care Excellence: Dementia: assessment, management and support for people living with dementia and their carers. NICE guideline [NG97]. Published: 20 June 2018
2. Diagnostic and Statistical Manual of Mental Disorders, 5th edition, Text Revision (DSM-5-TR). American Psychiatric Association Publishing, Washington, DC, 2022. pp 672-674
3. Nasreddine ZS, Phillips NA, Bédirian V, et al: The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment [published correction appears in J Am Geriatr Soc. 2019 Sep;67(9):1991. doi: 10.1111/jgs.15925]. *J Am Geriatr Soc* 53(4):695–699, 2005. doi:10.1111/j.1532-5415.2005.53221.x
4. Statement on use of apolipoprotein E testing for Alzheimer disease. American College of Medical Genetics/American Society of Human Genetics Working Group on ApoE and Alzheimer disease. *JAMA* 1995;274(20):1627-1629.
5. Fleisher AS, Pontecorvo MJ, Devous Sr MD, et al: Positron emission tomography imaging with [18F] flortaucipir and postmortem assessment of Alzheimer disease neuropathologic changes [published correction appears in *JAMA Neurol*. 2023 Aug 1;80(8):873. doi: 10.1001/jamaneurol.2023.1911]. *JAMA Neurol*. 77(7):829–839, 2020. doi:10.1001/jamaneurol.2020.0528

Treatment of Dementia

- Measures to ensure safety
- Provision of appropriate stimulation, activities, and cues for orientation
- Elimination of medications with sedating or anticholinergic effects
- Possibly cholinesterase inhibitors and memantine or anti-amyloid monoclonal antibody therapy
- Assistance for caregivers
- Arrangements for end-of-life care

Measures to ensure patient safety and to provide an appropriate environment are essential to treatment, as is caregiver assistance (1, 2). Several medications are available.

Patient safety

Occupational and physical therapists can evaluate the home for safety; the goals are to

- Prevent accidents (particularly falls)
- Manage behavior disorders
- Plan for change as dementia progresses

How well patients function in various settings (ie, kitchen, automobile) should be evaluated using simulations. If patients have deficits and remain in the same environment, protective measures (eg, hiding knives, disabling the stove, removing the car, confiscating car keys) may be required. Some states require physicians to notify the Department of Motor Vehicles of patients with dementia because at some point, such patients can no longer drive safely.

If patients wander, signal monitoring systems can be installed, or patients can be registered in a safe return program.

Ultimately, assistance (eg, housekeepers, home health aides) or a change of environment (living facilities without stairs, assisted-living facility, skilled nursing facility) may be indicated.

Environmental measures

Patients with mild to moderate dementia usually function best in familiar surroundings.

Whether at home or in an institution, the environment should be designed to help preserve feelings of self-control and personal dignity by providing the following:

- Frequent reinforcement of orientation
- A bright, cheerful, familiar environment
- Minimal new stimulation
- Regular, low-stress activities

Orientation can be reinforced by placing large calendars and clocks in the room and establishing a routine for daily activities; medical staff members can wear large name tags and repeatedly introduce themselves. Changes in surroundings, routines, or people should be explained to patients precisely and simply, omitting nonessential procedures. Patients require time to adjust and become familiar with the changes. Telling patients about what is going to happen (eg, about a bath or feeding) may avert resistance or violent reactions. Frequent visits by staff members and familiar people encourage patients to remain social.

Rooms should be reasonably bright and contain sensory stimuli (eg, radio, television, night-light) to help patients remain oriented and focus their attention. Quiet, dark, private rooms should be avoided.

Activities can help patients function better; those related to interests prior to the patient's dementia onset are good choices. Activities should be enjoyable, provide some stimulation, but not involve too many choices or challenges.

Exercise to reduce restlessness, improve balance, and maintain cardiovascular fitness and muscle tone should be done daily. Exercise can also help improve sleep and manage behavior disorders.

Occupational therapy and **music therapy** help maintain fine motor control and provide nonverbal stimulation.

Group therapy (eg, reminiscence therapy, socialization activities) may help maintain conversational and interpersonal skills.

Medications

Eliminating or limiting medications with central nervous system (CNS) activity often improves function. Sedating and anticholinergic medications, which tend to worsen dementia, should be avoided.

The **cholinesterase inhibitors** donepezil, rivastigmine, and galantamine are somewhat effective in improving cognitive function in patients with Alzheimer disease or dementia with Lewy bodies and may be useful in other forms of dementia. These medications inhibit acetylcholinesterase, increasing the acetylcholine level in the brain.

Memantine, an NMDA (*N*-methyl-D-aspartate) antagonist, may help slow the loss of cognitive function in patients with moderate to severe dementia and may be synergistic when used with a cholinesterase inhibitor.

While cholinesterase inhibitors and memantine are considered symptomatic treatment for Alzheimer disease and other dementia, disease modifying anti-amyloid monoclonal antibody treatment is also available (see Disease-modifying treatment for Alzheimer disease).

Medications to control behavior disorders (eg, antipsychotics) have been used.

Patients with dementia and signs of depression should be treated with nonanticholinergic antidepressants, preferably selective serotonin reuptake inhibitors (SSRIs).

Caregiver assistance

Immediate family members are largely responsible for care of a patient with dementia (see Family Caregiving for Older Adults). Nurses and social workers can teach caregivers how best to meet the patient's needs (eg, how to deal with daily care and handle financial issues); teaching should be ongoing. Other resources (eg, support groups, educational materials) are available.

Caregivers may experience substantial stress. Stress may be caused by worry about protecting the patient and by frustration, exhaustion, anger, and resentment from having to do so much to care for someone. Health care professionals should watch for early symptoms of caregiver stress and burnout and, when needed, suggest support services (eg, social worker, nutritionist, nurse, home health aide, and short-term patient stay in a nursing home for respite care).

If a patient with dementia has an unusual injury, the possibility of elder abuse should be investigated.

Treatment references

1. NICE: National Institute for Health and Care Excellence: Dementia: assessment, management and support for people living with dementia and their carers. NICE guideline [NG97]. Published: 20 June 2018
2. Alzheimer's Association: 2018 Alzheimer's Association Dementia Care Practice Recommendations. January 18, 2028. Accessed February 3, 2025.

Prognosis for Dementia

Dementia is usually progressive. However, progression rate varies widely and depends on the cause. Dementia shortens life expectancy, but survival estimates vary. In one prospective study in the United States, in patients with Alzheimer disease, median survival from initial diagnosis was 4.2 years for males and 5.7 years for females (1). In a meta-analysis involving over 5 million people followed for a minimum of 1 year following a dementia diagnosis, median survival was 4.8 years and median time to nursing home admission 3.3 years; survival was longer for women and somewhat longer for patients with Alzheimer disease, compared to other diagnoses (2).

End-of-life issues

Because insight and judgment deteriorate in patients with dementia, appointment of a family member, guardian, or lawyer to oversee finances may be necessary. Early in dementia, before the patient is incapacitated, the patient's wishes about care should be clarified, and financial and legal arrangements (eg, durable power of attorney, durable power of attorney for health care) should be made. When these documents are signed, the patient's capacity should be evaluated, and evaluation results recorded. Decisions about artificial feeding and treatment of acute disorders are best made before the need develops.

In advanced dementia, palliative measures may be more appropriate than highly aggressive interventions or hospital care.

Prognosis references

1. Larson EB, Shadlen MF, Wang L, et al: Survival after initial diagnosis of Alzheimer disease. *Ann Intern Med* 140(7):501–509, 2004. doi:10.7326/0003-4819-140-7-200404060-00008
2. Brück CC, Mooldijk SS, Kuiper LM, et al. Time to nursing home admission and death in people with dementia: systematic review and meta-analysis. *BMJ*388:e080636, 2025. Published 2025 Jan 8. doi:10.1136/bmj-2024-080636

Key Points

- **Dementia, unlike age-associated memory loss and mild cognitive impairment, causes cognitive impairments that interfere with daily functioning.**

- Be aware that family members may report sudden onset of symptoms only because they suddenly recognized gradually developing symptoms.
- Consider reversible causes of cognitive decline, such as structural brain disorders (eg, normal-pressure hydrocephalus, subdural hematoma), metabolic disorders (eg, hypothyroidism, vitamin B12 deficiency), medications, depression, and toxins (eg, lead).
- Do bedside mental status testing and, if necessary, formal neuropsychologic testing to confirm that cognitive function is impaired in ≥ 2 domains.
- Recommend or help arrange measures to maximize patient safety, to provide a familiar and comfortable environment for the patient, and to provide support for caregivers.
- Consider adjunctive medications, and recommend making end-of-life arrangements.

More Information

The following English-language resource may be useful. Please note that The Manual is not responsible for the content of this resource.

Alzheimer's Association

Drug Information for the Topic